

# Equation Sheet: Optics

Intro Physics II • Lectures 28–33 • Knight ch. 33–34

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**fundamental** = know cold (your starting points); **derived** = a line or two from the fundamentals.

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## Symbols and constants

|                                       |                                     |                                            |
|---------------------------------------|-------------------------------------|--------------------------------------------|
| $n$ index of refraction ( $n = c/v$ ) | $s, s'$ object / image distance (m) | $\lambda$ wavelength (m)                   |
| $\theta$ angle from the normal / axis | $f$ focal length (m)                | $d$ slit separation or grating spacing (m) |
| $\theta_c$ critical angle             | $m$ image magnification             | $a$ single-slit width (m)                  |
| $R$ radius of curvature (m)           | $L$ slit-to-screen distance (m)     | $c$ $3.00 \times 10^8$ m/s                 |

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## 1. Reflection, refraction & total internal reflection (Knight 34.1–34.4)

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### fundamental

Law of reflection  $\theta_{\text{incidence}} = \theta_{\text{reflection}}$

Index of refraction  $n = \frac{c}{v}$   
(light slows in a medium;  $n \geq 1$ ;  $\lambda_n = \lambda_{\text{vac}}/n$ )

Snell's law  $n_1 \sin \theta_1 = n_2 \sin \theta_2$   
(toward the normal entering a higher  $n$ ; away  
entering a lower  $n$ )

### derived

Critical angle  $\sin \theta_c = \frac{n_2}{n_1}$   
(only for  $n_1 > n_2$ ; beyond  $\theta_c$ , total internal  
reflection)

Normal incidence ( $\theta_1 = 0$ )  $\theta_2 = 0$  (no bending, any  $n_1, n_2$ )

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## 2. Thin lenses & mirrors (Knight 34.5–34.7)

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### fundamental

Thin-lens / mirror equation  $\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}$

Magnification  $m = -\frac{s'}{s} = \frac{h'}{h}$   
( $m < 0$  inverted,  $m > 0$  upright;  $|m| > 1$   
enlarged)

### derived

Spherical mirror focal length  $f = \frac{R}{2}$

Sign conventions  $f > 0$ : converging lens / concave mirror;  $f < 0$ :  
( $s' > 0$  real image;  $s' < 0$  virtual) diverging / convex

Object inside the focal point ( $s < f$ ,  
converging) virtual, upright, enlarged ( $m > 1$ )

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### 3. Interference of light (Knight 33.1–33.3)

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#### fundamental

Constructive / destructive interference  
(path-length difference  $\Delta r$ )  $\Delta r = m\lambda$  (bright),  $\Delta r = \left(m + \frac{1}{2}\right)\lambda$  (dark)

Double slit — bright fringes at angle  $\theta$   $d \sin \theta = m\lambda$ ,  $m = 0, 1, 2, \dots$

#### derived

Fringe spacing on the screen (small angles)  
(wider spacing for larger  $\lambda$  or smaller  $d$ )  $\Delta y = \frac{\lambda L}{d}$

Position of the  $m$ th bright fringe  $y_m = \frac{m\lambda L}{d}$

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### 4. Diffraction gratings & single-slit diffraction (Knight 33.4–33.6)

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#### fundamental

Grating — bright maxima  
(same form as the double slit, but the maxima are sharp and narrow;  
 $d = 1/(\text{lines per unit length})$ )  $d \sin \theta = m\lambda$

Single slit — *dark* fringes  
( $a$  is the slit width; the central bright band spans  $\pm\lambda/a$ )  $a \sin \theta = m\lambda$ ,  $m = 1, 2, 3, \dots$

#### derived

Highest observable grating order  $m_{\max} \leq \frac{d}{\lambda}$  ( $\sin \theta \leq 1$ )

White light through a grating each order spreads into a spectrum; red bends most

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### 5. Polarization (Knight 33.7)

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#### fundamental

Malus's law — polarized light through an analyzer  $I = I_0 \cos^2 \theta$   
( $\theta$  between the light's polarization and the axis;  
zero when crossed)

#### derived

Unpolarized light through one polarizer  $I = \frac{1}{2}I_0$   
(output is polarized along the axis; independent of the axis angle)

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